



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS
PAPER 3 THEORY

6032/3

JUNE 2020 SESSION

2 hours 30 minutes

Additional materials:
Answer paper
Electronic calculator

TIME 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **four** questions.

Question 1 is compulsory.

Answer any other **three** from the remaining questions.

Write your answers on the separate answer paper provided.
If you use more than one sheet of paper, fasten the sheets together.
All working for numerical answers must be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.
You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 16 printed pages.

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3 (a) Define *capacitance* and the *Farad*. [2]

(b) When a $10\ \mu\text{F}$ capacitor C_1 is connected across a $7.2\ \text{V}$ dc source, $72\ \mu\text{C}$ of charge is deposited on its plates.

The capacitor is now placed in a vacuum without losing any charge as shown in Fig. 3.1.

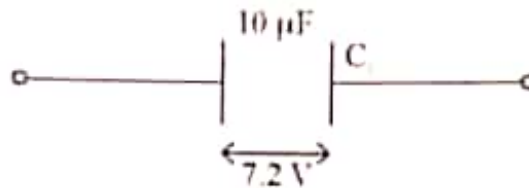


Fig. 3.1

- (i) Calculate the energy stored on the capacitor C_1 .
- (ii) A second uncharged $12\ \mu\text{F}$ capacitor C_2 is now connected across C_1 as shown in Fig. 3.2.

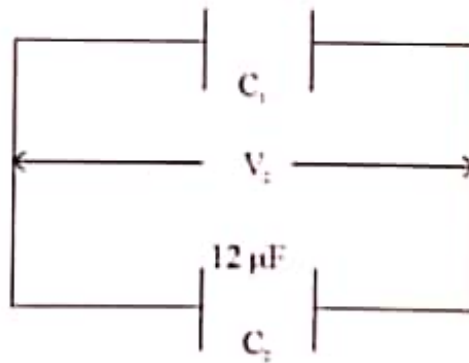


Fig. 3.2

Determine the

1. effective capacitance due to C_1 and C_2 ,
2. p.d. V_2 across the combination,
3. energy stored in the combination.

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$\frac{1}{C_{\text{total}}}$ [6]

$$C = \frac{W}{V}$$

5 (a) Distinguish between emission spectra and absorption spectra. [2]

(b) (i) Explain the terms

1. *wave-particle duality.*
2. *de Broglie's wavelength.*

(ii) In an electron gun, electrons are accelerated from the cathode by a p.d. of 225 V.

1. Calculate the maximum speed of an electron towards the anode.
2. Determine the de Broglie wavelength associated with an electron travelling at the speed calculated in (b)(ii) 1.
3. Identify the part of the electromagnetic spectrum to which the wavelength in (b)(ii) 2 belongs.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

[7]

(c) (i) Deuterium (${}^2_1\text{H}$) and Tritium (${}^3_1\text{H}$) are isotopes of hydrogen. They react to give a helium nucleus and a neutron.

Write a nuclear equation to show the reaction.

(ii) Estimate the energy released during the nuclear reaction in (c)(i).

$$\left[\begin{array}{l} m({}^2_1\text{H}) = 2.01410u, m({}^3_1\text{H}) = 3.0160u, m({}^4_2\text{He}) = 4.00260u \\ m({}^1_0\text{n}) = 1.00866u \end{array} \right] \quad \text{c.c.f.} \quad [4]$$

(d) A milling company suspects that part of a batch of flour packs is contaminated with a radioisotope of strontium – 90 (Sr – 90). A test is to be carried out using a Geiger-Muller tube.

- (i) Describe the procedure that must be taken to accurately measure the activity of a radioactive sample.
- (ii) Calculate the time taken for the activity to fall to 30% of the initial value.

[Decay constant of Sr – 90 is $7.75 \times 10^{-10} \text{ s}^{-1}$]

- (iii) Fig. 2.2 shows a monochromatic parallel beam of light directed towards point A on a vertical wall.

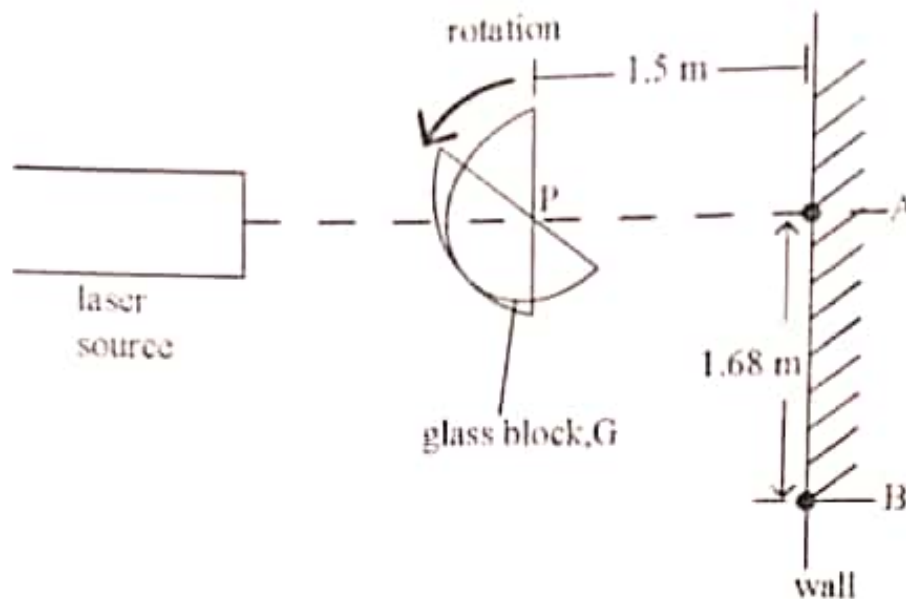


Fig. 2.2

A semi-circular glass block, **G**, is placed symmetrically across the path of light a distance 1.5 m from the wall producing a bright spot on point **A**. When the block, **G**, is rotated about point **P**, the bright spot moves from **A** to **B** and immediately disappears after point **B**.

Explain why

1. the bright spot moves from **A** to **B** as **G** is slowly rotated anticlockwise,
2. the light disappears as the block **G** is rotated further.

- (iv) Determine the refractive index of the glass block **G**.
- (v) Predict whether distance **AB** will be shorter or longer if glass of a higher refractive index is used.

[11]

- (c) Oil has a linear attenuation (absorption) coefficient of 0.59 cm^{-1} for a particular incident X-ray beam.

Calculate the depth required to reduce the intensity of the X-ray beam to 0.0027 of its incident intensity.

[3]

(b) (i) Define *Young Modulus*.

(ii) Fig. 4.2 shows a copper wire ABC of length 1.4 m and diameter 3.0 mm fixed between two supports.

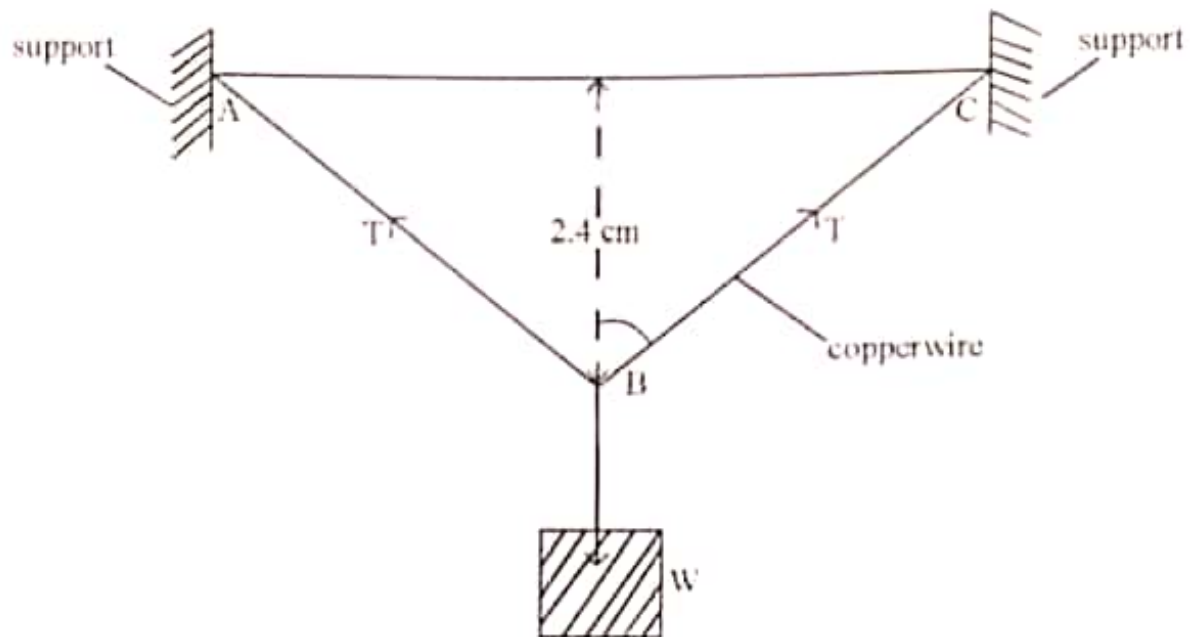


Fig. 4.2

A weight of 98 N is hung from the midpoint of the wire, B, such that the point sags 2.4 cm below the horizontal line.

Calculate for the wire, the

1. extension when the weight is hung,
2. tension,
3. Young Modulus.

(iii) State any **two** assumptions made in (b)(ii).

[11]

(c) (i) State the *Zeroth Law of Thermodynamics*.

(ii) Explain why

1. when a person is holding a piece of metal rod whilst the other end of the rod is in contact with ice, the end he is holding finally becomes cold as well,
2. a thermocouple records a higher value of e.m.f. when placed near a red hot metal than when placed near the human body.

[5]

- (d) A sample object consists of 4 voxels. The object is scanned from 4 different directions. The readings for each individual voxel are summed up and the result is shown in Fig. 2.3.

23	50
32	35

Fig. 2.3

The total background reading is 20.

Determine the initial pattern of the pixels in the voxels.

[2]

(iii) Suggest a way of disposing the contaminated flour packs safely.

[7]

(e) (i) Define *signal attenuation*.

(ii) Fig. 5.1 shows a communication channel with two repeater units 100 km apart. The gain of the amplifier is 37.2 dB.

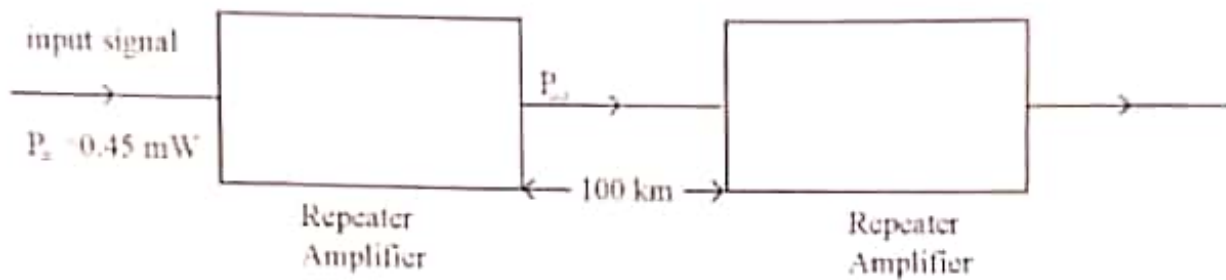


Fig. 5.1

Calculate the

1. power output of the amplifier,
2. attenuation per unit length of the cable.

[5]

- (c) Fig. 3.3 shows a negatively charged solid metal sphere placed in a vacuum.

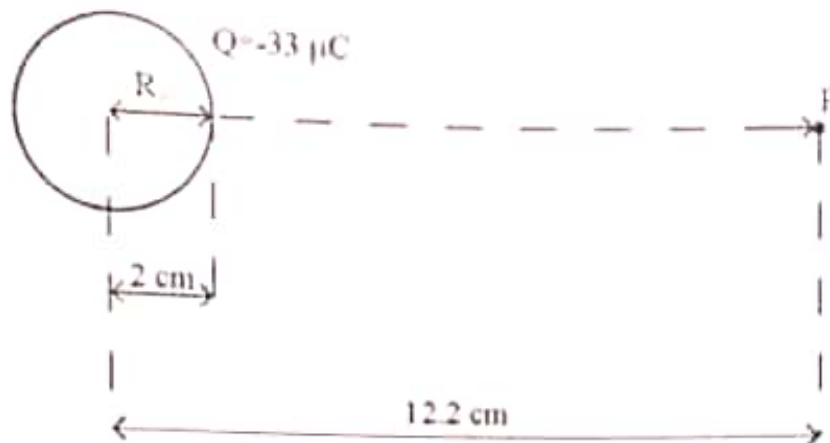


Fig. 3.3

- Calculate the magnitude of the electric field strength at point P.
- Sketch the variation of electric potential V with distance r from the centre of the sphere to point P.
- State any **two** differences in the general characteristics between the field of an isolated point charge and an isolated planet.

[6]

- (d) Fig. 3.4 shows a potential divider circuit.

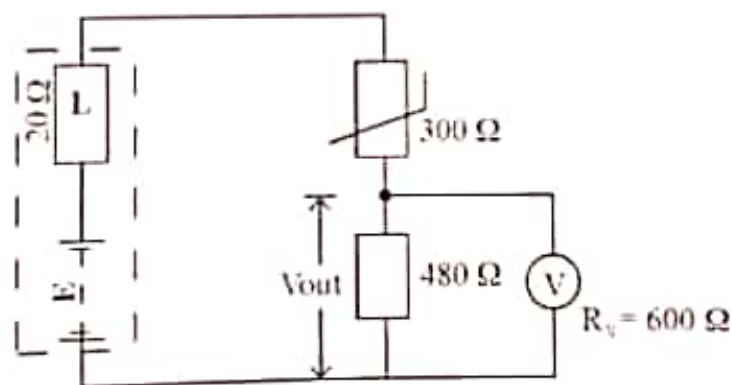


Fig. 3.4

Calculate the

- magnitude of V_{out} ignoring the value of R_v ,
- actual voltmeter reading.

[5]

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(d) A milling company suspects that part of a batch of flour packs is contaminated with a radioisotope of strontium – 90 (Sr – 90). A test is to be carried out using a Geiger-Muller tube.

- (i) Describe the procedure that must be taken to accurately measure the activity of a radioactive sample.
- (ii) Calculate the time taken for the activity to fall to 30% of the initial value.

[Decay constant of Sr – 90 is $7.75 \times 10^{-10} \text{ s}^{-1}$]

2

- (a) (i) Explain the term *phase difference*.
- (ii) Fig. 2.1 shows a displacement-time graph of a body of mass 0.2 kg undergoing simple harmonic motion.

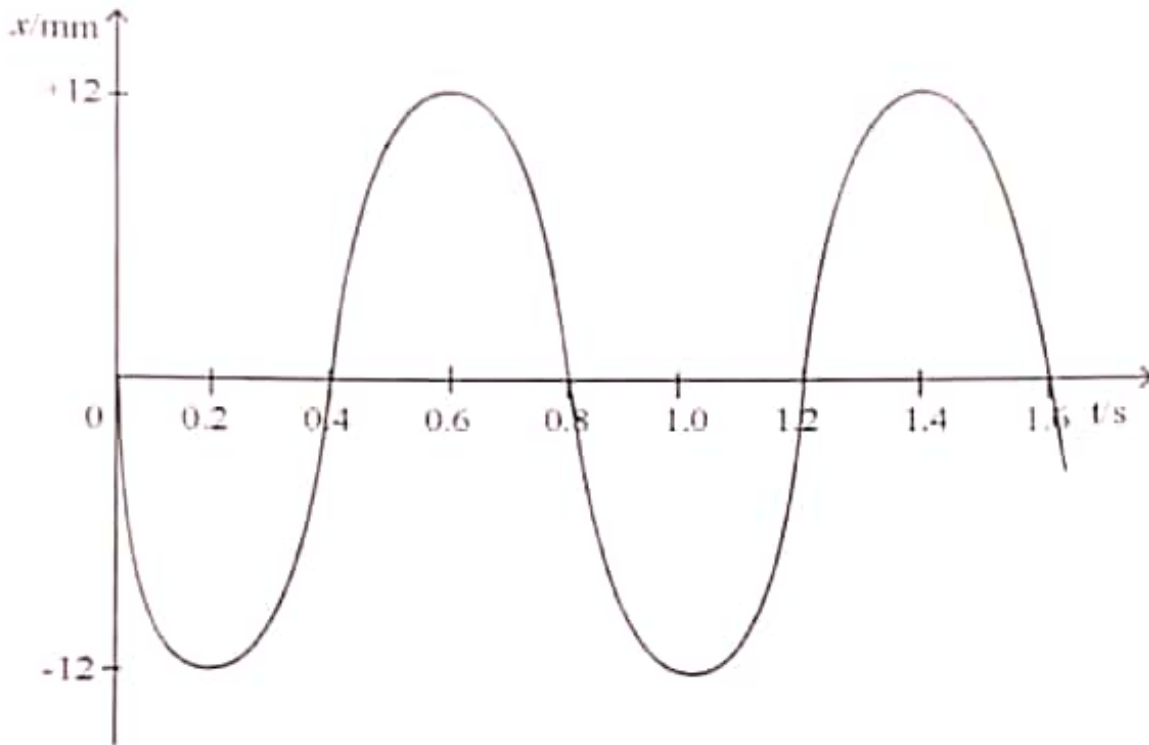


Fig. 2.1

1. Calculate the angular velocity of the oscillation.
2. Write the equation of the velocity of the oscillation in the form $x = x_0 \sin \omega t$.
3. Sketch the velocity-displacement graph of the oscillation.
4. Hence or otherwise deduce the maximum energy of the oscillation.

[9]

- (b) (i) Define *total internal reflection*.
- (ii) Explain why the optical density of the core of an optical fibre must be greater than that of the cladding.

V =

The body rolls down from rest and reaches the bottom of the plane.

- (i) Calculate the
1. acceleration of the body as it moves down the plane,
 2. velocity of the body on reaching the bottom of the plane.

The body hits the bottom and rolls up to point **B** with the same acceleration as in (b)(i).

- (ii) Calculate the velocity of the body at the bottom after rolling down **OB**.
- (iii) Suggest forms in which energy is lost at point **O**.

[8]

- (c) (i) Use Newton's first law to explain why a passenger in a lift feels lighter when the lift stops while ascending.
- (ii) The system in Fig. 1.2 is in equilibrium. The maximum force that the string can withstand is 100 N.

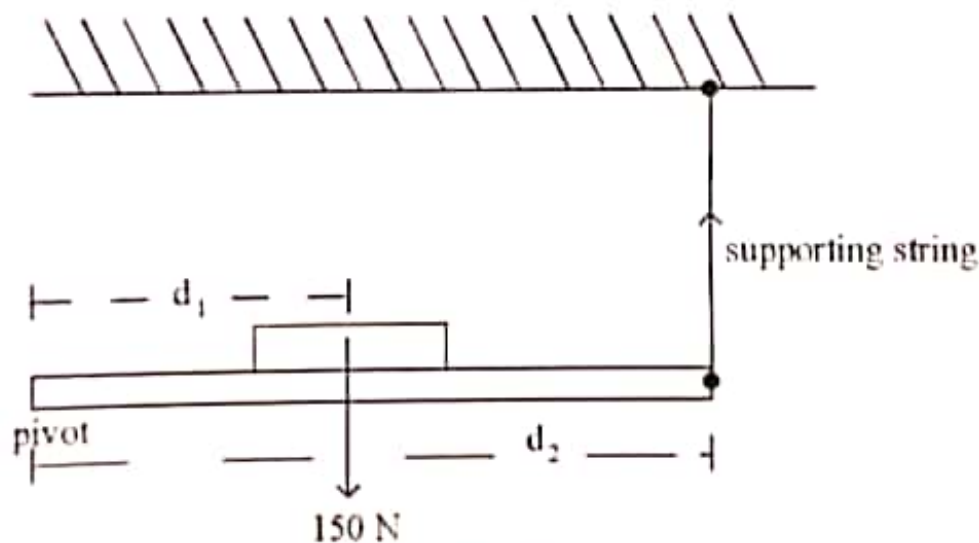


Fig. 1.2

Use the principle of moments to determine the maximum possible value of the ratio $\frac{d_2}{d_1}$.

[4]

- 4 (a) (i) State two separate ways of increasing the internal energy of an ideal gas.
- (ii) Fig. 4.1 shows a $P-V$ graph for an ideal gas in a perfectly insulated cylinder.

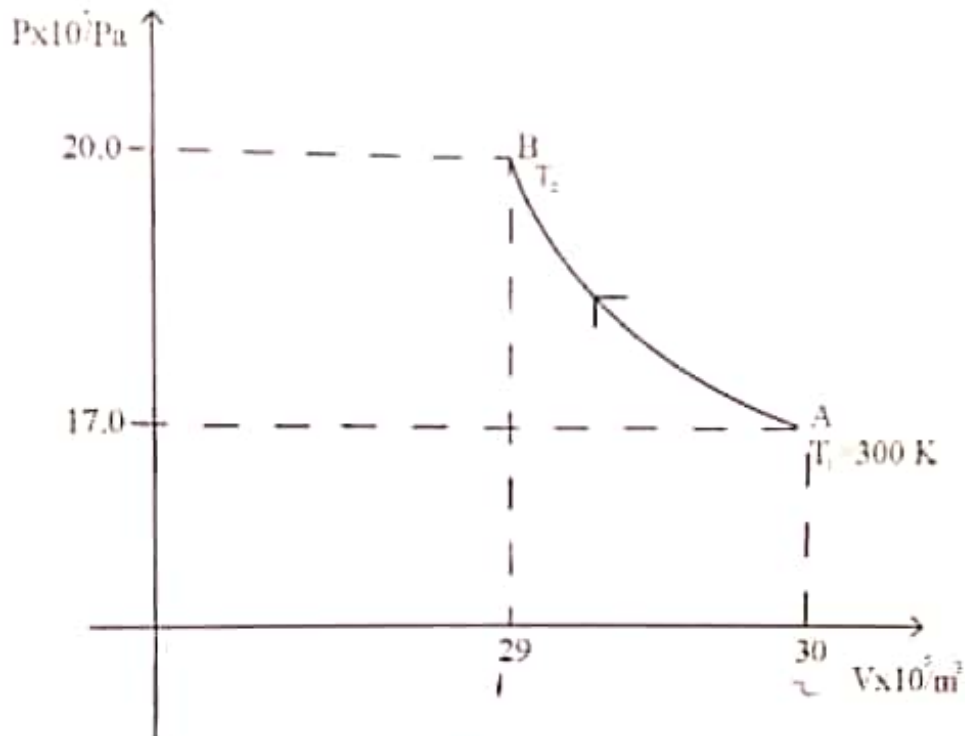


Fig. 4.1

Using Fig. 4.1

1. state a reason why, in Section A to B, work is said to be done on the system,
 2. determine the temperature of the gas at point B,
 3. determine the change in internal energy between A and B if the work done on the system is approximately 17.6 J.
- (iii) Explain why in practical situations, the temperature at B is lower than the value calculated in (ii)2.

[9]

- (e) Fig. 3.5 shows the circuit of a Schmitt trigger with positive feedback.

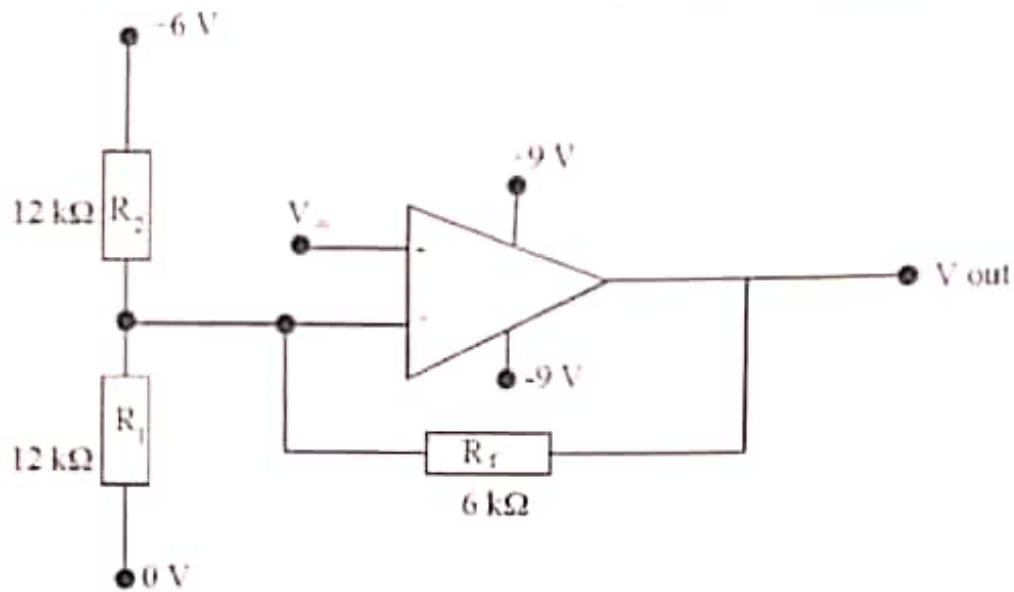


Fig. 3.5

- (i) Describe the behavior of the output voltage V_{out} as V_{in} varies from 0 to +6 V then down to -6 V.
- (ii) Explain **one** practical application of a Schmitt trigger.
- (iii) Describe any **two** positive impacts of digital electronic systems on organisational security.

[6]

- (d) (i) Define *gravitational potential* at a point in a gravitational field.
- (ii) Fig. 1.3 shows a body of mass 1 000 kg being moved along the path shown in a uniform gravitational field.

Gravitational
potential

MJ Kg⁻¹

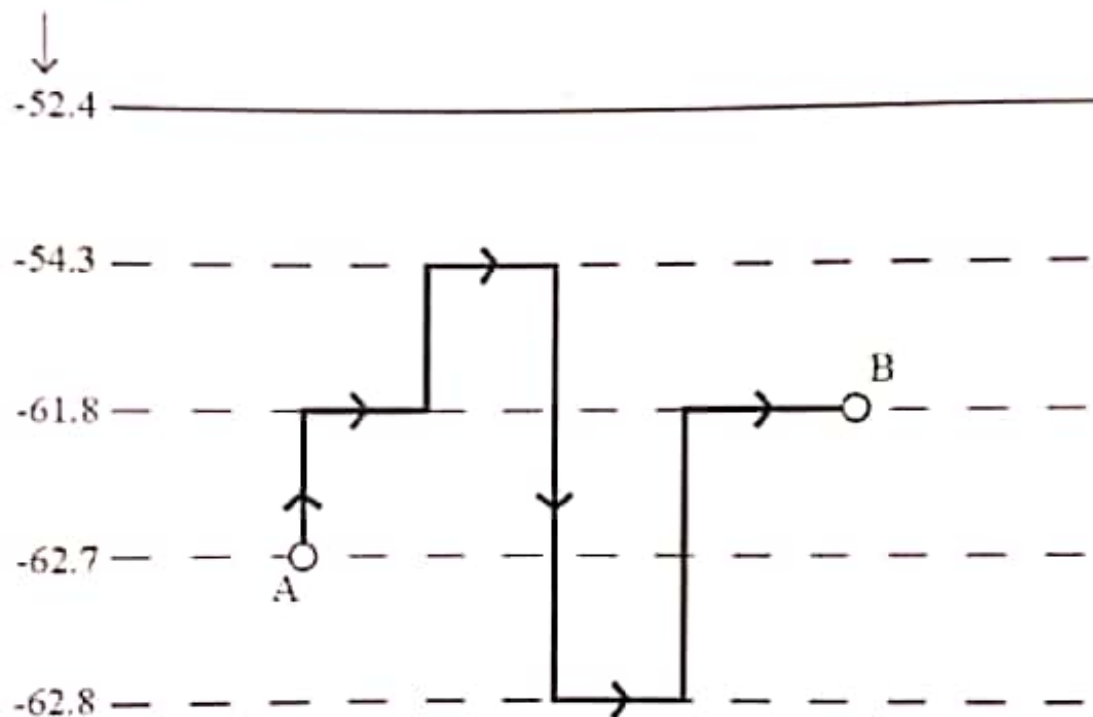


Fig. 1.3

Calculate the workdone in moving the body along the path shown.

- (iii) The gravitational field strength, g , near the surface of the earth is given a value 9.81 Nkg^{-1} .

Suggest why in reality the value varies.

[8]